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Flooding in Jakarta
Towards a blue city with improved water management

Sunday, 27 January 2002, a large flood swept down on Jakarta and inundated several parts of the city.[1] From the evening of 27 January to the morning of 28 January rain came streaming down, and the dike south of Jakarta broke. The pungent black water, with a hefty cargo of garbage, poured onto the main roads to Bogor, Kramat Jati and East Jakarta. In North Jakarta, in Kelurahan Pejagalan, Kecamatan Penjaringan, the flooding or *banjir* hit at midnight and continued until five o'clock in the morning, reaching levels as high as 20 cm. Even harder hit was Kelurahan Kapuk Muara, inundated with 70 cm of water.[2]

In West Jakarta the flood invaded houses and main thoroughfares. At Kelurahan Tanjung Duren Utara the waters rose 120 cm after the dike containing the Sekretaris and Gendong Rivers gave way, while the water at Tomang Barat Reservoir could no longer be controlled. River water spread over the housing estates and urban kampongs of West Jakarta. Daan Mogot Street and the housing in its vicinity were the first to be reached by the giant flood. Cars had to be evacuated before being overwhelmed by the deluge, and all furniture on the ground floor of two-storey houses had to be moved

1 This article is based on personal experience with floods in Jakarta, interviews with experts, and the study of historical material, newspapers, and official reports.
2 'Hujan lebat guyur Jakarta sejumlah permukiman tergenang', *Suara Pembaruan*, 28-1-2003.

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to the upper floor. Sometimes furniture had to be perched precariously on the roofs of buildings. The housing area along the West Banjir Canal turned into a large swimming pool filled with dirty, stinking water from the waterway, as there was no let-up in the rain for half a day. The park-like villa housing estate in the vicinity of Daan Mogot Street was flooded to a depth of one metre. The people who lived on this luxurious estate had to flee their houses since the floodwaters stayed there for weeks. The flood also ravaged Kelurahan Bambu Utara (Kecamatan Pal Merah) where more than 385 houses were standing in 15 cm of water. A number of streets in West Jakarta, such as Pesing (Kecamatan Grogol Petamburan), Kedoya Street (Kecamatan Kebon Jeruk), Kelapa Dua Street, and streets in the vicinity of Green Garden Duri Kepa Estate, suffered water levels 30 cm high. In South Jakarta the flood devoured Paku Buwono Street in Kebayoran Baru. It inundated the area to a depth of 30 cm. In East Jakarta the Cipinang River overflowed the whole of Kelurahan Cipinang Besar to 50 cm. Meanwhile, the dike at Kalibaru 21 km along the road to Bogor burst and the highway was covered with water half a metre deep. This caused a huge traffic jam. The water also flooded the kampongs behind the dike. Water one metre deep afflicted Serdang (Kecamatan Kemayoran) in Central Jakarta.

During the whole day of Monday, 28 January, thousands of houses in Jakarta were inundated as rain continued to barrel down. On the other side of the city, at Cililitan and Kampung Melayu, the water reached two metres. Houses were flooded and all the inhabitants had to seek refuge on rooftops. At Pejaten a landslide on the banks of Ciliwung River propelled four houses into the flood, killing six people.

So far no evacuation had taken place. The inhabitants of Jakarta thought that the flood had already reached its climax and would soon subside. People still went to work, and in spite of everything urban transportation continued to function. The city government took no action until Monday evening. Sutiyoso, Governor of Jakarta, believed that it was not necessary to declare the highest level of emergency.[3]

In East Jakarta the flood had seriously worsened. Cipinang River overflowed and more houses were inundated. A luxury estate, Cipinang Indah, was flooded to a depth of two metres. Many of the houses were hurriedly abandoned by their inhabitants with not enough time to save their possessions. A consulting firm in the area lost all its computers, books, maps, and project data as the house was hit by a flash flood early on the morning of 29 January.

The situation deteriorated when young people began demanding money from passers-by. These were groups of young gangsters who forced people to hand over their cash, an unusual occurrence in Jakarta.

[3] 'Banjir genangi Jakarta ribuan penduduk mengungsi', *Kompas*, 29-1-2003.

On 30 January at four o'clock in the morning, Suratno, a *soto* hawker in Dewi Sartika Street, was awakened by his wife when the five kilograms of noodles he had recently bought at the market was already afloat in the water that had unexpectedly entered the ground floor of their temporary two-by-two-metre house. Suratno hurried to take his small television upstairs because the water level was rising fast. The whole family was in a state of panic trying to save its possessions from the flood. Then suddenly the electricity went out and darkness prevailed everywhere.[4]

By now, the flood had become a serious problem, as more than three-quarters of Jakarta was under water. The city was like a giant Titan losing control when attacked by water. Traffic was at a standstill; for example, there was no movement for more than ten hours at the Kebon Jeruk Junction in West Jakarta. People had to turn off their engines and could not do anything in the middle of the traffic jam; the cars were surrounded by water like ships on the sea. Public transportation, which is normally abundant in the streets and stops everywhere, grew very scarce at the end of January and the beginning of February. Many inhabitants had to walk home, many of them for kilometres. In East Jakarta a group of men hijacked a minibus to take them to Pulo Gadung.

The water that had first flooded only a few spots in the city, by 30 January had covered almost the whole of Jakarta. Only Menteng, Pondok Indah, and Pantai Indah Kapuk remained unaffected. More than 40,000 people were forced to flee their houses. Traffic jams occurred along Daan Mogot Street, Cengkareng and Grogol. The alternative traffic route through Semanan also became blocked. Similar traffic jams occurred at Puri Indah, Kembangan and Meruya.[5]

Those who had already set off for the office had to return home when they noticed that the lines of cars in the traffic jam were not moving. This was the result of the inundation of the main road to a depth of 80 cm, so that cars could no longer continue because several of them had already been caught by the water.

The Cisadane River overflowed along Daan Mogot. Many vehicles were trapped on the street and could not move. An enormous traffic jam also blocked the toll road to Merak where thousands of vehicles had come to a halt for kilometres, meeting those coming from Kebon Jeruk. They were surrounded by floodwaters as high as 50 cm and could not budge an inch.

Another traffic jam occurred in East Jakarta. The Molek River at Pondok Gede overflowed the street to a depth of one metre. Vehicles were stuck waiting for the water to recede.

4 'Kisah pilu dari daerah banjir', *Penabur*, 2-1-2002.
5 'Banjir makin meluas', *Kompas*, 30-1-2003.

Not only did the flood create traffic jams, it wrought havoc on train and air services. At the Soekarno-Hatta Airport about 80% of flights were delayed. These were mainly domestic flights. A few international flights, which took off in the morning, were also behind schedule.

Those who were trapped in traffic jams and had enough money preferred to sleep in hotels. However, finding a room in a hotel was not easy, since most were full of well-to-do refugees whose houses were inundated. Hotels offered a temporary place to avoid the flood, but as the flood grew higher, their basement car-parks were inundated. In the end, when the generators were also flooded, the electricity was shut off and the more affluent people of Jakarta found themselves with the same problems as other inhabitants.[6]

The flood scotched the nightlife of Jakarta when people had to be evacuated from hotels. Electricity was cut off and lifts did not move. The red-light districts at Mangga Besar and in Gajah Mada Street no longer offered their usual entertainment services. Jakarta as a symbol of modern Indonesian life was hard hit. Nobody idealized Jakarta any longer as a place where dreams come true, as a destination for young people in search of a job. The flood, which in several places such as Kampung Melayu and Cililitan reached a depth of over six metres, traumatized inhabitants.

Because of its unique numerical composition, 02022002, 2 February 2002 had been predicted to be a lucky day. Hence, on that day many couples had planned to get married. Of course, a wedding reception is a very important occasion to show off one's affluence. To achieve the maximum effect, there are firms specializing in creating a wedding programme that includes luxurious cars, an expensive venue for the reception, delicious food, and a high-class hotel for the bride and bridegroom. Those who had paid for a wedding reception on that day were disappointed. A marriage event at a luxury hotel in Kuningan received only a few guests in contrast to the crowds of people who had been invited. One disappointed bride was reduced to tears realizing that this once-in-a-lifetime event had gone down the drain. As a consequence of the flooding, Jakarta's high life was completely extinguished.

The flooded areas were not only those occupied by low-income groups, they also included the Kelapa Gading district inhabited by more prosperous groups. Fighting broke out between the residents of Sunter and Kelapa Gading when they disagreed on whether the floodgate on the Sunter River should be opened or not. Kelapa Gading and Sunter are two large housing estates just across the Yos Sudarso Bypass. If the floodgate were closed the water would overflow the Kelapa Gading Estate and the housing in Sunter would remain safe. An inhabitant of Kelapa Gading claimed that the floodgate had been closed by a person from Sunter. The car factory, Astra, located on the Sunter

6 'Perkawinan batal; Penghuni hotel mengungsi', *Kompas*, 3-2-2003.

side, hired soldiers to guard the closed floodgate. Were this gate to have been opened, Astra would have lost quintillions of rupiahs because its new cars would have been inundated. As a consequence of the floodgate being closed, the flood inexorably spread to Kelapa Gading. This luxury estate was transformed into a lake. The inhabitants of Kelapa Gading went to the floodgate to ask the gate-keeper to open it halfway. After a long debate the floodgate was opened a few centimetres and then the opening mechanism cut off. Kelapa Gading was inundated for a week. In order to go out of their houses, people had to pay dearly, as much as Rp 100,000, to get onto a raft.[7]

The current system of dealing with flooding has not changed since the division of the Ciliwung River into two channels: the West Banjir Canal traversing the southern and western parts of the city, and the main stream of the Ciliwung River, called Inner City Ciliwung River, flowing through Kenari Street, Raden Saleh Street, Kali Pasir Kwitang, and Istiqal, where it was divided again into two smaller rivers, the one that passes Pasar Baru and Gunung Sahari to Ancol, and the other that flows by Gajah Mada Street and Kota. Manggarai is the place where the floodgate is located, guarded by soldiers. In 2002 the floodgate to the West Banjir Canal was open so that the floodwaters flowed into West Jakarta. Meanwhile the floodgate to the Inner City Ciliwung was closed. If this floodgate had been open, the presidential palace, Thamrin Street, and Kota would have been inundated. Since this floodgate was closed, the flood in the eastern parts of the city including Cililitan and Kampung Melayu was very high. People went to the Manggarai floodgate and wanted to open it, but they had to face soldiers who had instructions to shoot anyone who touched the gate.

The city government of Jakarta had not anticipated large parts of the city being submerged, although as early as 3 January 2002, the Board of Meteorology and Geophysics had issued a warning that from mid-January to early February extensive flooding could occur in Jakarta. During this period the spring tide would be at its maximum, coinciding with the highest average rainfall. Despite the warnings, the city government considered this a regular event that did not warrant special attention. The equipment it had at its disposal was obsolete and could not cope with a flood of this magnitude.[8] Not until the early morning of Saturday, 2 February, did the central government announce the highest state of emergency. The floodgate on the Inner City Ciliwung River was opened. The grounds of the presidential palace, the National Monument, and Kota, forming the central administrative and economic heart of the country, were then flooded.[9]

7 'Perkawinan batal Perusahaan Listrik Negara (PLN), Penghuni hotel mengungsi', *Kompas*, 3-2-2002.

8 'Pertengahan Januari diperkirakan Jakarta mulai banjir', *Kompas*, 3-1-2003.

9 'Jakarta terendam total', *Kompas*, 2-2-2002.

The water brought economic activity in the capital to a standstill. All offices including banks were closed, and streets jammed with traffic were a common scene on the periphery. The golden triangle of M.H. Thamrin, Jendral Sudirman, Gatot Subroto, and H.R. Rasuna Said was very quiet. A similar calm could be observed in Kota, Hayam Wuruk, Gajah Mada, Roxy, Tomang and Cempaka, which are normally busy and congested. Meanwhile, the three-day-old flood had caused the State Electric Company (PLN) a loss of Rp 15-16 billion since it was forced to shut off 120 electricity sub-stations supplying 400 megawatts of power. Some trains were put out of order and the route from Pasar Minggu, Manggarai and Gondangdia went no farther than Tebet Station. The flood also affected Tanah Abang Station, which could not cope with the outpouring from the West Banjir Canal. Spill-over forced the cancellation of a series of electric train services, while other lines experienced delays. Cancellation of trains stranded thousands of commuters waiting at various stations.[10]

According to the Ciliwung-Cisadane River Basin Development Project (2002) about 15% of the Jabotabek[11] area of 100 km^2 was flooded. Local bottlenecks (mostly man-made structures) were believed to be responsible for most of the flooding of higher ground in the city. At the lower levels closer to the sea most of the inundations were caused by low hydraulic gradients in combination with profile constriction and narrow bed width. An overview of the areas inundated within DKI[12] Jakarta during the 2002 flood is presented in Figure 1. The areas most affected were situated in northwestern and northeastern Jakarta, with water depths of over 1.5 metres. These areas consisted of the downstream reaches of the Cengkareng Floodway System, the West Banjir Canal System, and the downstream reaches of the East Banjir Canal System. Among the other areas affected were the remaining coastal area of North Jakarta and areas further upstream along the Ciliwung and Pesanggrahan Rivers. Flooding also occurred outside DKI boundaries. The number of fatalities caused by the flooding of 2002 is estimated at about eighty. Direct damage is estimated at Rp 5.4 trillion, whereas indirect losses are estimated at Rp 4.5 trillion (NEDECO 2002).

In the post-flood period, the people of Jakarta were afflicted by many health problems. Diarrhoea, flu and skin diseases were especially common. Everywhere people cleared the mud from their houses. Most families lost valuable items such as furniture and vehicles. According to Mari'i Mohamad,

[10] *Kompas*, 30-1-2003. Flood declared national disaster, economic activities paralysed, 8 dead, 80% of flights suffered delay.

[11] The acronym Jabotabek stands for the cities of Jakarta, Bogor, Tangerang and Bekasi. It comprises the Greater Jakarta area and functions as a planning region.

[12] DKI stands for Daerah Khusus Indonesia and means that Jakarta is a special administrative region at provincial level.

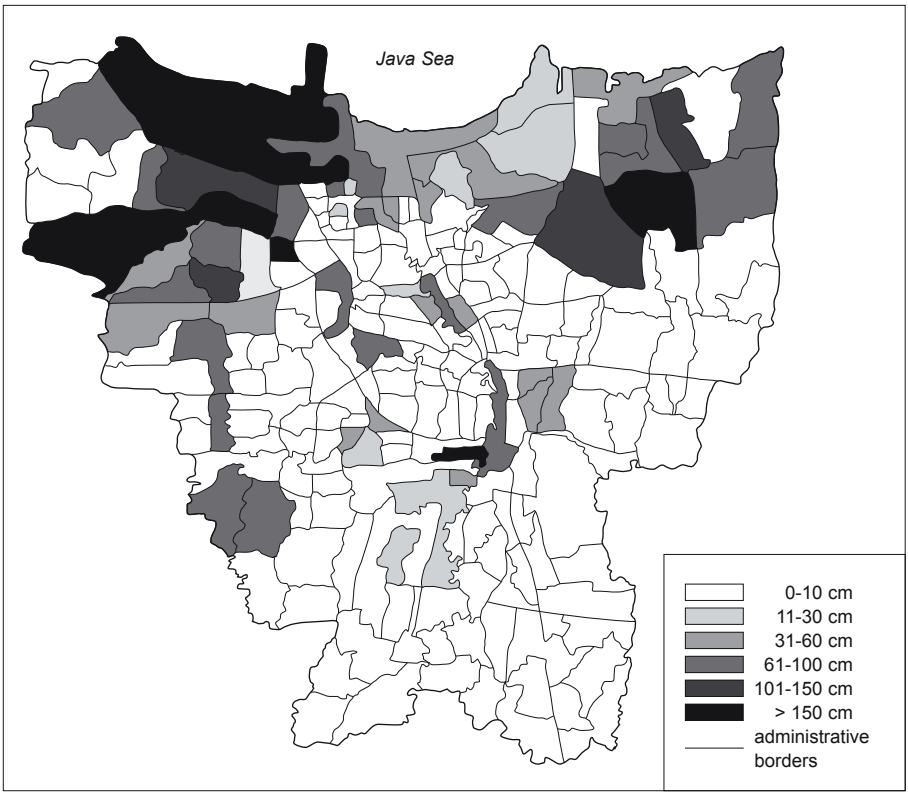

Figure 1. Areas inundated during the 2002 flood (based on Rijkswaterstaat et al. 2003)

the head of Red Cross Indonesia, conditions were most difficult for poor people, as they did not have the money to buy new beds or to repair their houses.[13]

The flood had a serious impact on the property market. The criteria for buying a house are no longer limited to a strategic location close to a school, place of work, and shopping mall, but also include low risk of flooding. The price of places like Kelapa Gading, which was famous for its strategic position in Jakarta, have declined as it was inundated for a week. According to Kalim and Sulaeman (2002), 68.4% of the owners of houses that were inundated were prompted to sell their houses, 24.6% did not want to sell their houses, and another 6.8% wanted to wait and see. Unfortunately, those who wanted to sell their houses could not find buyers. Since the flood, Kelapa Gading has suffered from declining property values.

[13] 'Banjir, kemiskinan dan korupsi', *Tempo*, 25-2-2002.

After the flood, sand bags were placed along the street as protection. People raised the floors of their houses, often without the required permission from the city planning board. Developers raised estate ground levels by several metres. The new Mangga Dua Mall at Gunung Sahari is situated two metres higher than the level of the surrounding houses. But even here the street level was heightened by two metres, so that only three metres of space was left under the railway viaduct that crosses the street. All these reactions show that people in the capital cared mainly about the safety of their own property without paying attention to the behaviour of others. The well-off, who managed to pay for the construction of new, higher floors, did not bother about the situation of their less fortunate neighbours. As a consequence, the poor, who did not have the money to raise their floor levels, will suffer worse flooding of their houses in the future. People's reactions show how large is the gap between rich and poor.

After the disaster, the central government planned to spend Rp 15 quintillion through the Badan Perencanaan Pembangunan Nasional (BAPPENAS, National Planning Board) to overcome the problem of flooding in Jakarta within ten years. Hence, each year there is Rp 1.5 quintillion available for the city government to run the project. The reaction of Jakarta public opinion was negative. People believe this sum of money will go towards further corruption by the city government and cause new scandals. A coalition of several NGOs against this plan protested to the head of the National Planning Board.

Not content with this, the public criticized the city government's budget for the year 2002. The majority of the amount of Rp 9 quintillion that had been agreed on by the City Council was actually assigned to routine costs, such as salaries and other routine expenditures. Flood problems came in a poor second. The public agitated for the budget to be revised such that at least 30% would be used to tackle flood problems.[14]

In this kind of situation, the main public figure targeted for criticism was of course Sutiyoso, Governor of Jakarta. He responded to the flood too late. People reproached him for ignoring the Badan Meteorologi dan Geofisika's (BMG, Meteorology and Geophysics Board) warning, so that when the flood came, the city government was at a loss about how to respond. As many as 29 NGOs and other community organizations stated that Governor Sutiyoso had failed to lead the city properly and had made wrong decisions in attempting to solve the flood problems. Indignantly they asked the Assembly of Jakarta to dismiss Sutiyoso from his function as governor.[15] The main director of change in the capital experienced a difficult moment. To avoid further attacks, Sutiyoso took the initiative and destroyed his own

[14] 'DKI harus revisi RAPBD', *Kompas*, 4-2-2002.
[15] 'DPRD didesak berhentikan Sutiyoso', *Kompas*, 6-2-2002.

villa at Puncak, a resort area that has been developed rapidly in the last two decades, a situation which some believe to be one of the causes of the flooding in Jakarta (*Suara Merdeka*, 14 February 2003).

The second person who became a target of criticism was Ciputra. This famous developer has built many housing areas including Pantai Indah Kapuk, which was spared the waters, while its surroundings were inundated.[16]

The third person criticized was Megawati, then President of Indonesia. She was blamed for being very late to react. In fact she visited victims on 31 January, four days after the flood struck the city (*Suara Pembaruan*, 1 February 2002). The President and her team encountered protests from the flood victims they visited.

The flood of 2002 shows that in the Reformasi era people have become more critical of the government. They openly blame the Governor and his administration for mismanagement and corruption. The flood was considered the result of decades-long mismanagement of the city's water management infrastructure. Although the flooding stifled city life for 'merely' a week, it continued to cause trauma for more than a month.

Now, several years after the flood, it is interesting to note that the tragedy of 2002 seems to be almost forgotten, or perhaps memory of it has been suppressed. It was only after the flood had really taken hold that bureaucrats finally reacted, but notwithstanding ongoing projects, it is still not clear to the general public what decisions have been taken and what plans have been realized to protect the city from another, probably worse, rainy season. Exacerbating this distrust, the city government even plans to reclaim 2,700 ha of coastline and predicts that they will thereby earn Rp 12 quintillion from investors. As the flood was caused by a lack of water-catchment areas, such reclamation without compensatory measures would probably result in even wider flooding in Jakarta.[17]

Nowadays the people of Jakarta give the impression of having forgotten the tragedy. The Rp 15 quintillion plan is no longer mentioned in the press. NGOs are busy with other matters that are more sensational than floods in a floodless period. The names of housing estates that fell victim to the water are rarely mentioned. People have stopped thinking about it. The garbage that was blamed as one of the factors causing flooding has now ceased to be a topic of debate, and garbage is still floating everywhere in the rivers and canals of the city. Boats for the evacuation of victims have been stored away. The Puncak area is still considered an attractive place to build villas. There is no more public debate about the problem of water-catchment areas.

[16] 'Pantai Indah Kapuk penyebab banjir', *Media Indonesia*, 10-2-2002.

[17] 'Pemprov DKI kejar keuntungan Rp 12 trilyun', *Kompas*, 4 December 2003 (Pemprov is the acronym of *permerintah propinsi,* provincial government).

The flood is like a monster patiently waiting to seize its chance because of the idleness of city inhabitants and their officials; at a certain moment it will return to destroy city life, worse than before. The events of January and February 2002 show that the flooding problems of Greater Jakarta have grown progressively worse. The high intensity and long duration of the rainfall inundated vast areas, even those that normally remain unaffected. Many people had to flee their homes, contagions spread, power plants shut down, and the toll roads and main thoroughfares were closed. The 2002 disaster claimed a large number of human casualties, disruption of city life for over a month, and significant direct and indirect damage to property and the economy.

Location

To understand these extreme circumstances, the position of Jakarta in the coastal area of Java has to be taken into account. When the city was founded on the site of the small port town of Kalapa, later changing its name to Jayakarta, Jacatra, Batavia, and eventually Jakarta, people could not possibly have imagined that this small settlement on the 5,000-year-old alluvial fanlike plain, composed of the debris of the Salak, Pangrango and Gedeh volcanoes, would develop into a huge city and become the core of a mega-urban region with tens of millions of inhabitants. This site, on a seashore that is slowly expanding because of sediment carried by the rivers and taking the shape of a large bay in which lie numerous small islands, was positioned just above sea level. That is why this lowland area (depicted in Figure 2) has always been subject to regular natural flooding by the waterways which cut through the plain, such as the Cisadane, Angke, Ciliwung, Bekasi and Citarum Rivers. In the hot season the rate of discharge of these rivers is low and urban canals are mostly a dirty, muddy trickle carrying smelly sediment. But during the wet season, when these rivers carry the bulk of their sediment – silt that in the course of time has grown in quantity because of upstream human-induced erosion-promoting developments – the risk of flooding is high. The often disastrous consequences have been aggravated over the centuries by rapid urbanization leading to extension of the built-up environment and consequently greater numbers of human beings liable to be affected, and to human action and inaction that have led to obstruction and deterioration of the drainage system, seriously reducing its discharge capacity.

In the past few decades, the population of Jabotabek has exploded from 2.7 million in 1960, to 6 million in 1980, and to 11.5 million in 1995 (*Urbanization prospects* 1995). While the city expanded and developed rapidly, the existing drainage system was neglected. Uncontrolled urbanization, poor spatial planning, a lack of maintenance, and the government's inertia in implementing

improvements to the drainage system have led to the current situation in which most drains and rivers do not have sufficient discharge capacity to cope with increasing peak flows. Consequently, flooding events have become more frequent and more extensive.

In the Soeharto era and then under the new government, the city's need for additional space has resulted in mushrooming housing and commercial developments in the Bogor and Puncak areas. Forests and paddy fields have been quickly replaced by asphalt pavements and concrete buildings, immediately reducing the area's water-retention capacity. More rainwater is diverted faster and more directly into the nearest stream or river, thereby increasing river peak flows and discharges. As a result, a lower amount of rainfall, which previously would have been partly retained and would not have caused any problems downstream, may now cause flooding.

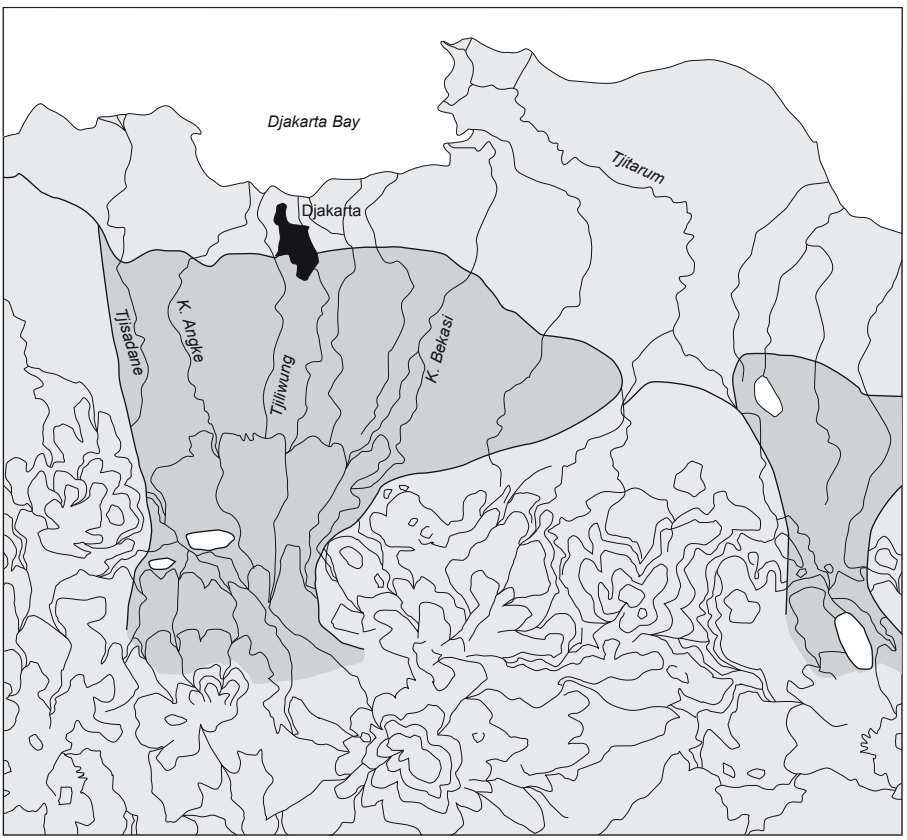

Figure 2. Outline map of the Bay of Jakarta with the hinterland, scale 1:1,000,000. Dark grey area: alluvial plane (based on Verstappen 1953:8).

At the same time, because of deforestation and increasing peak flows, the amount of sediment flowing towards and into rivers has increased. Sediment transportation capacity is generally high in the upstream sections of the river and does not cause any problems there. In the lower reaches, however, where there are urbanized areas, the sediment settles, decreasing the flow cross-section. Without proper maintenance (dredging) of river beds, the flow cross-sections will continue to decrease and rivers and drains will overflow their banks more easily. The same situation obtains in the river outlets, some of which are heavily silted up. Small hydraulic gradients in the lower reaches of the rivers and drains lead to low flow velocities, especially during high tides. At present, parts of North Jakarta inundate easily during spring tides and most of the water has to be discharged at lower tides (larger hydraulic gradient). Nevertheless, sediment deposits at river outlets continue to hamper smooth disgorgement into the Java Sea, creating backwater effects.

The problem is exacerbated by most drains and rivers being clogged with garbage, which many people dump into rivers and drains. According to the World Bank (*Environment and development* 1994), only 40% of trash and garbage is collected, and 30% is believed to get blown or washed away or is dumped into rivers and drains. Garbage tends to heap up at outlets, along riverbanks, against bridge abutments, in front of floodgates and other man-made structures. This narrows the flow cross-section of waterways and can create backwater effects.

Even without garbage, many of the man-made structures, such as floodgates and bridges, cause a narrowing of flow cross-sections and are known to create backwater effects that can cause flooding upstream. This may be caused by a lack of maintenance, or the structure may simply be outdated, no longer adequate to meet current requirements for discharge capacity. At some locations along rivers and drains, peak flows may be hampered by provisional wooden houses and other small structures on the river banks, built by many of Jakarta's poor who have settled on river banks and estuaries.

In North Jakarta flood risks are rising because of the effect of land subsidence, caused by uncontrolled and unlimited deep groundwater extraction. In some areas of North Jakarta rates of 6 cm per year have been measured. It is expected that land subsidence will continue in the near future and parts of North Jakarta will slowly sink below sea level. Flood risks will be heightened by subsiding river dikes and by greater inundation depths and floodwater velocities during flooding, leading to more damage. At the same time, bridges and other drain- and river-spanning infrastructure are subsiding. At some locations there is hardly any clearance left between the bridge deck level and floodwater levels. When this happens, the bridge becomes a major obstacle to floodwater discharge.

Besides rainfall and low ground level, urbanization is clearly one of the

main factors in the flooding problem. It is the cause of the loss of water-retention capacity, even in upstream areas where deforestation is occurring, and lies at the root of gradual land subsidence in certain far northern parts of the city. Lack of capacity and maintenance of drainage infrastructure are also important. This is compounded by socio-cultural factors such as policy implementation, solid waste dumping, and management. In reports by experts, many concrete proposals for improvement of the drainage system are presented. A comprehensive approach is proposed that, in addition to water-retention areas, also includes a flood monitoring and early warning system as inundations will clearly continue to occur in the foreseeable future.

So, both the description of the 2002 flood and the depiction of the geographic and urbanization situations of Jakarta make it clear that the capital of Indonesia has a problem to deal with. To foster a comprehensive approach to Jakarta flooding, we will now describe the water conditions of the city, presenting a historical overview, an evaluation of current infrastructure works, and present administrative views. The core question to be answered is: Why has the persistent problem of regular flooding in Jakarta never been solved?

History

After the founding of Batavia in 1619 a system of canals was gradually created, very much like that of Amsterdam at the time. The soil dug up was used to raise the level of land for construction purposes. In 1633 complaints were already being aired about the stench when the canals were lying dry, and in 1665 it was pointed out that the water sometimes rose higher than the level of the town. By 1670 some streets were inundated at the time of the spring tide. In his comprehensive work on the situation in Old Batavia, De Haan reports extensively on canals and flood conditions.[18] The differences between the tides, although generally quite small, sometimes led to flooding at spring tide and to the drying up of some of the canals at low tide. Problems with sandbanks at the mouth of the Ciliwung and with the slow but inexorable extension of the coastline were also noted in the early days of Batavia. By means of a palisade wall, efforts were made – without much success – to protect the castle against the extension of the coastline, but finally a canal had to be dug in the newly accrued terrain. The sandbank at the mouth of the river was dredged with the help of nets and dredging mills powered by horses.[19]

De Haan (1922) mentions a number of causes and effects of flooding in the town of Old Batavia and offers some solutions. As the root causes, he adduces the low level of the land and the small difference between the tides.

[18] De Haan 1922, I:236-8, 248-9, 252-3, 385, 401.
[19] These mills were dismantled in 1709.

The cutting down of trees upstream on the Ciliwung led to higher water levels in this river and raised its sediment level. The crumbling of the canal banks also augmented the quantity of sediment transported. Another factor was water distribution over extended *sawah* areas, which led to the dispersal of water over larger areas, rendering it unmanageable. He also mentioned the use of the river and canals for the removal of solid waste. The sugar mills were named as major culprits because of the large quantities of organic wastes dumped in the river before it entered Batavia.

Yet the main cause was the sinking of mud at places where the current was less strong, or where it came to a standstill because of debris, mainly at the inner corners of the canals and near the coastline where there was only a limited flow. In these quiet waters, mud had more time to form sandbanks that could grow into small mud pools and even end up becoming islands that were to grow and change place in the course of time.

The upshot of the situation was of course regular inundations and muddy, stinking canals as the water became scarcer in the dry season. In the lower parts of town it was impossible to construct sewage systems, as at high tide sewage water would flow into the surface water.

Plenty of solutions were tried. Efforts were made to stop the silting up by dredging with nets, with dredging mills, and with manual (forced) labour wielding small baskets. The pier in the bay of Jakarta was extended, but the sandbank at the mouth of the river only appeared to be pushed forward. The lack of efficaciousness of the palisade wall around the castle and the need for construction of a canal have already been mentioned. Strenuous efforts were made to prevent waste being dumped in canals. In 1740 the river was even diverted through different canals, which it entered at a right angle to stop the strong current. All sorts of waterworks were constructed, including solid canal banks, sluices, and dams.

In 1725 a dam forced the waters of the Ciliwung through the Westerse Vaart, but in times of flooding the river tried to retake its old bed and the dam had to be opened to prevent its complete destruction. This was the first in a series of flood canals intended to bridle the Ciliwung. All the early canals aimed at controlling flooding, such as the Bacherachtsgracht and the Mookervaart, were oriented westwards. Only relatively recently has an East Banjir Canal, now under construction, been considered.

In later centuries, other authors have also recounted the flood problems of Batavia. Abeyasekere (1987:70-1, 74) remarks on the high frequency of flooding in the nineteenth century, which often failed to rouse the government from its inertia. The author suggests that inhabitants and officials got used to the occurrence of floods and came to a fatalist acceptance of them. She argues that the authorities only felt obliged to undertake large drainage works when there had been particularly bad floods. This was the case in the 1870s. Some

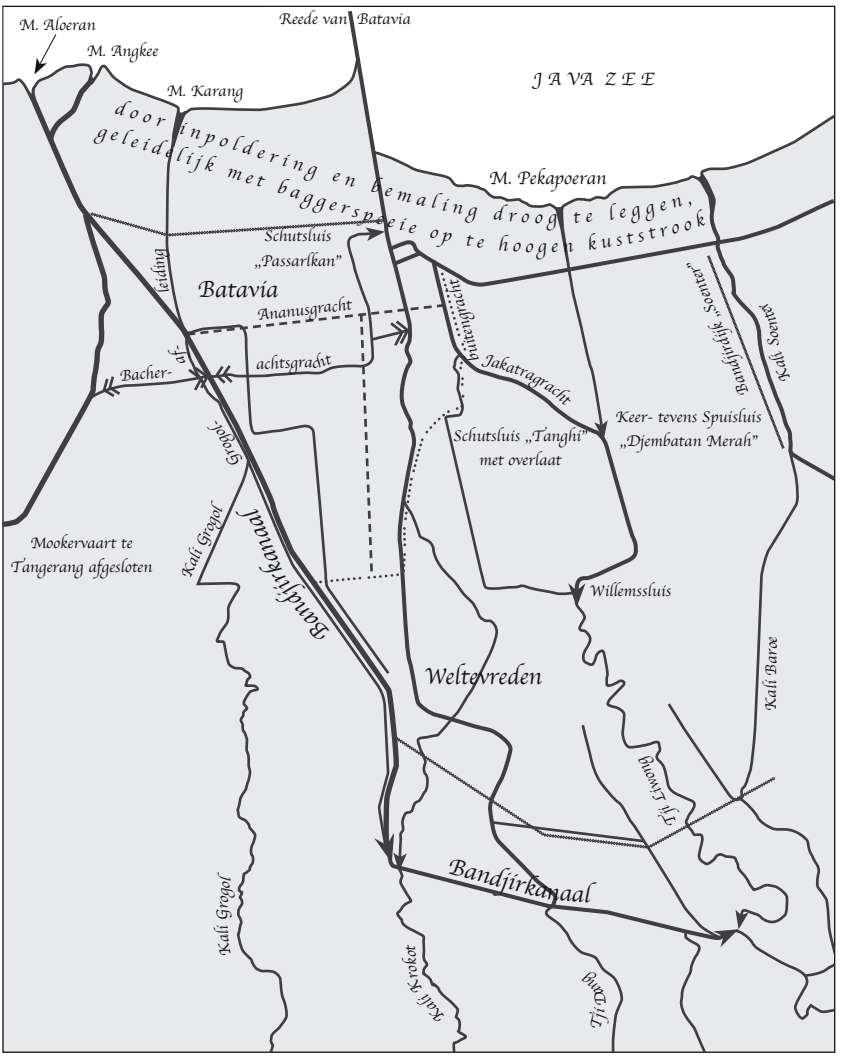

Figure 3. The planned situation in Batavia after construction of the Banjir Canal (based on Van Breen 1917:2b)

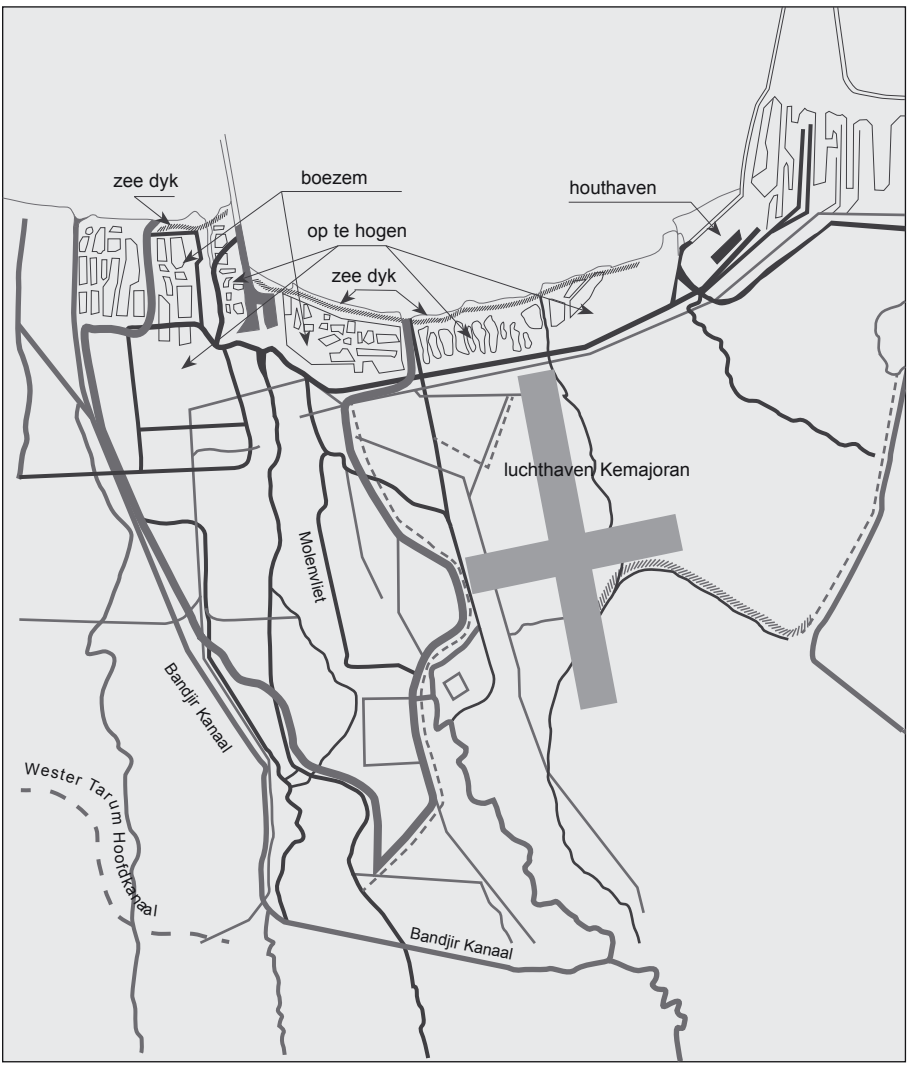

Figure 4. Plan for Batavia and Tanjung Priok (based on Van Blommestein 1948: Plaat I)

of the canals in the lower town were then filled in and new ones were dug further upstream to divert floodwaters.

In the twentieth century the proposal of H. van Breen (1917) for a western flood canal, the Banjir Canal, was realized. In Figure 3 the course of the projected canal is shown. It diverts the water of the Ciliwung south of Menteng to both the west and the north, finally entering the sea near Muara Angke. It is interesting to see that along the north coast a polder was proposed, its level to be heightened by dredged mud. Van Breen also proposed a variety of smaller works to improve hygiene and drainage. Several of these were implemented, as reported by Eggink (1930:65-7) in his commemorative publication on the 25th anniversary of the municipality of Batavia.

In the 1940s W.J. van Blommestein drew up a comprehensive plan for the irrigation and drainage of the whole of West Java. It consisted of dams and lakes in the Citarum River for white electricity (hydroelectric power), new irrigation fields near the town of Cilacap, and the construction of a dike along the sea between Batavia and Tanjung Priok to develop a polder with several basins on the site of the coastal fish ponds. Van Blommestein also listed several other elements essential to better water supply and drainage in Batavia, such as the deepening of the canals, pumping installations, sluices, and a drinking water installation.

This historical review shows that in the course of time many solutions for the flooding problems of the city were proposed and indeed sometimes realized. Most of these aimed at westward drainage through new canals, protecting both the old city and the new colonial centre of Menteng. Regular maintenance was also assumed, but was generally quite unsatisfactory. The idea of a system of basins in a polder along the coast was brought forward, but not realized. These plans and projects had to follow the tremendous expansion of the city that took place over the centuries and clearly lagged behind it. A definitive answer has not yet been found and, as was shown, regular flooding has been a persistent scourge for the inhabitants.

Jakarta

On the basis of the Jakarta Master Plan for 1965-1985, Sethuraman (1976) presents an evaluation of the drainage and flood control conditions in the 1960s. He points out that:

(a) nearly all the rivers flowing in the city, in combination with the drains network, are now functioning in a very unsatisfactory manner;
(b) most of the rivers and drains have become shallow owing to the practice of dumping refuse and waste directly into them, and because of inadequate maintenance and dredging;

(c) most estuaries have already become shallow through a continuous silting-up process which greatly hampers the flow of water;
(d) almost all of the water constructions are seriously damaged and operating inefficiently;
(e) many areas are periodically submerged as a result of ineffective road drainage; and
(f) there is insufficient open ground, and total water-retention capacity is thus diminished. (Sethuraman 1976:38.)

In 1965 the Departemen Pekerjaan Umum (DPU, Office of Public Works) initiated a long-term project:

> which foresaw the excavation of 3.4 million cubic metres of earth to provide water reservoirs, the construction of 89 kilometres of canals, and the dredging of 750,000 cubic metres of earth in rivers and estuaries, as well as the construction of pumps, bridges, and flood control gates and the relocation of residents from an area of 478 hectares. It is reported that two-fifths of the project was completed in the first three years (1965-67). In 1967 expenditure on the project amounted to Rp 28.5 million. (Sethuraman 1976:39.)

Besides these anti-flood measures, Jakarta planning was developed in a wider context and given more structure in the framework of the Jabotabek planning region. The idea was that the growth of Jakarta had to be curbed in favour of other cities in the planning region, namely Bogor, Tangerang and Bekasi, in order to protect the capital city from the overwhelming problems resulting from rapid population increase. Under Governor Ali Sadikin, an attempt was even made to close the city, though without much success. Flood protection could easily be integrated into this city-planning model as substantial green areas between the centres were projected. Yet the goal was still to find a better solution to recurrent flooding and in the early 1970s Dutch consultants (NEDECO) proposed an eastern flood canal. The plans for this East Banjir Canal (EBC) were set out in more detail in the 1990s by Japanese consultants. The canal is intended to lead the water of the five rivers east of the Ciliwung to the sea, and it is scheduled to be finished in 2010, though some delay is expected.

In the meantime, planning for Pantura (Pantai Utara, or the North Coast) was developed, prompted by problems with the drinking water supply resulting from heavy pumping of groundwater.[20] This practice, which is

[20] The planning concept of PANTURA should not be confused with the large-scale plans for land reclamation that also bear the name PANTURA. Between 1990 and 1999 the government of Jakarta approved this massive land reclamation project. The project consists of the reclamation of 2,700 ha of land from the sea in Jakarta Bay, stretching along 32 km of present coastline from Banten Province past DKI Jakarta up to West Java Province. Early in 2003 the Minister of the Environment released a decree rejecting the project for environmental reasons.

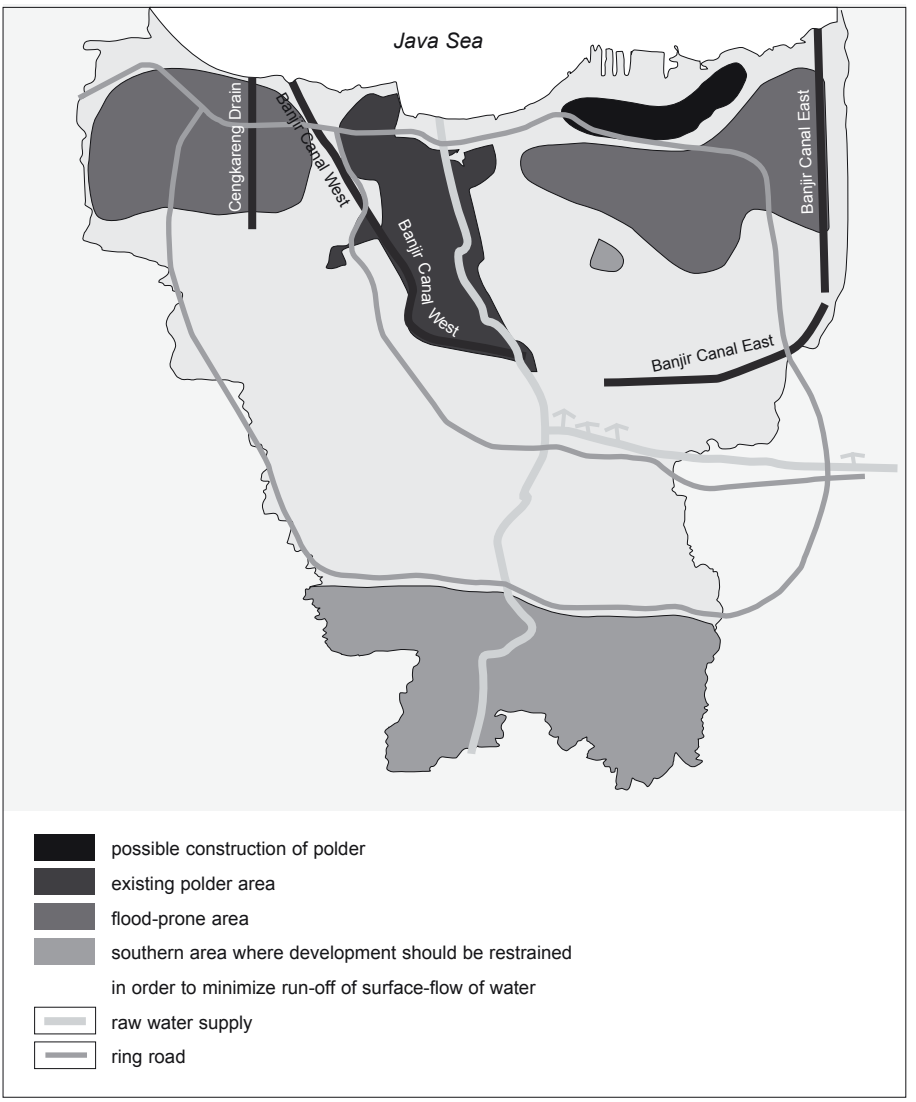

Figure 5. Jakarta's 2005 flood control and drainage plan
(based on *Jakarta 2005* 1991:54)

very common in Jakarta, with households pumping water from uppermost groundwater layers and hotels and industries from lower layers, causes the emptying of these layers and the intrusion of seawater so that nowadays the northern part of the city can no longer use groundwater for drinking. This situation has led to the idea that Jakarta should not extend to the south, further aggravating the groundwater problem, but rather should turn to the east and the west. These ideas related to Jabotabek and Pantura are clearly indicated in the current master plan.

The Jakarta master plan of 1984 for the period 1985-2005 devotes a great deal of attention to flood control and drainage. The main targets are improvement in the flow of water, storage of rainwater for cleansing the canals during the dry season, and integration of macro, semi-macro and micro drainage networks. The flood control system is divided into three parts: central zone, west zone, and east zone. In the central zone Depok Dam and improvement of the existing polder system including new pumps are emphasized; in the west zone the completion of the Cengkareng Flood Canal takes priority; and in the east zone the completion of the upstream part of the East Banjir Canal is planned for 2005 (*Jakarta 2005* 1991:23). The action plan lists several priorities, such as construction of Depok Dam, continued use of artificial lakes in upstream areas for flood control, improvement of coordination between the various agencies responsible for ensuring that clean rivers are kept free of garbage, household sewage, and industrial waste, and restriction on urban construction in southern parts of the city (*Jakarta 2005* 1991:55).

The report also stresses that the handling of micro drainage and sub-micro drainage for flood control requires a very large investment. Cooperation with central government agencies is deemed essential and therefore this should be continuously stimulated (*Jakarta 2005* 1991:69).

Figure 5 presents the flood control and drainage plan as indicated in the Jakarta 2005 Master Plan. The flood-prone areas in the western and eastern parts of Jakarta appear substantial. In fact, these are large expansion areas between the city and Cengkareng International Airport and between the city and Tanjung Priok Harbour: mangrove swamps as well as *sawah* areas. The solution for both sides of the city has clearly been sought in the respective flood canals: Cengkareng and East Banjir Canal (the latter still under construction). The central city zone, including the old town and Menteng, constitutes a polder and is protected by the West Banjir Canal. A new polder is proposed for the area directly behind Tanjung Priok Harbour. The southern zone of the city is marked for retarded urban development in order to minimize surface flow of water to the built-up parts of the city. The implementation of these plans received strong impetus in the wake of the 2002 flooding.

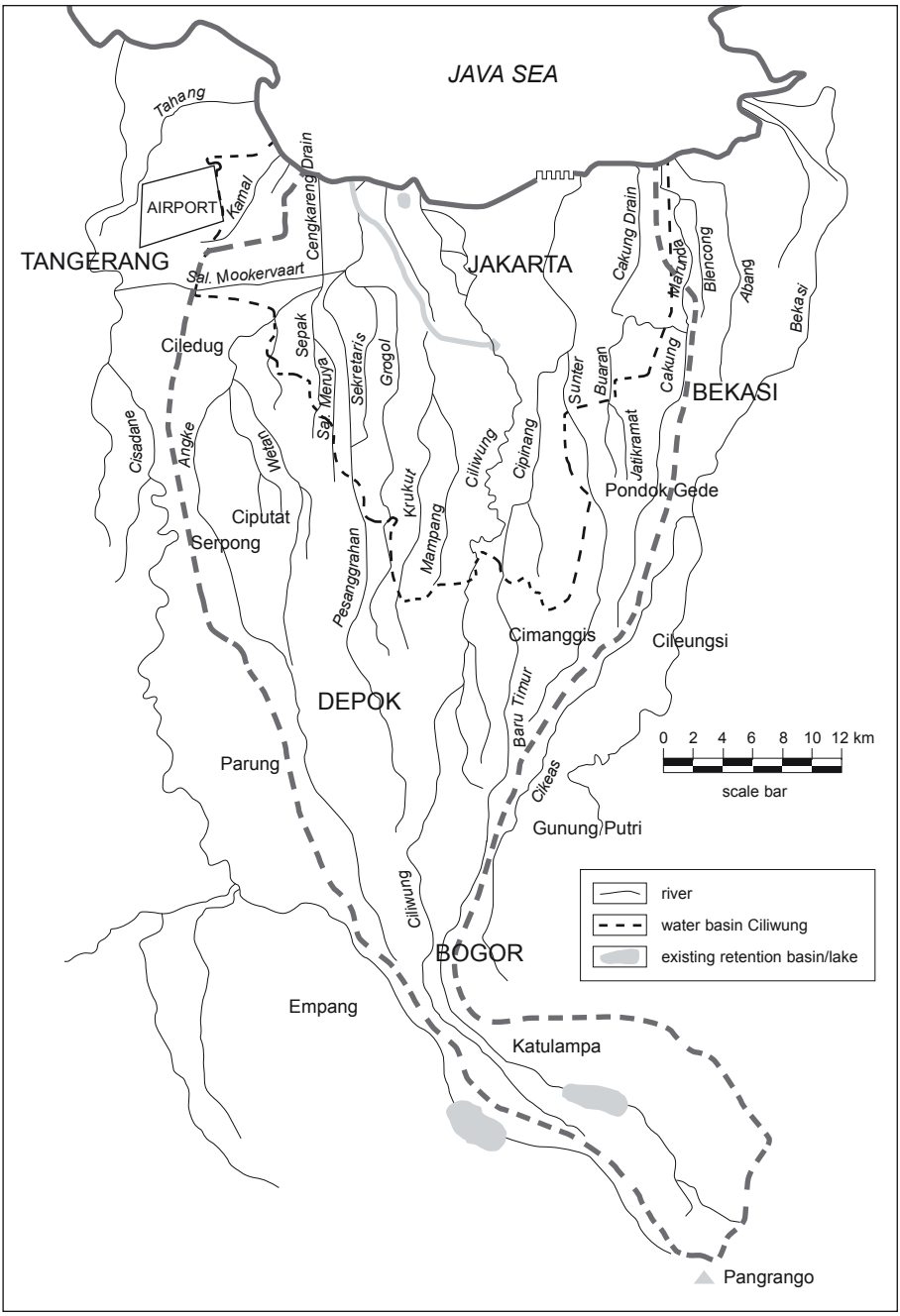

Figure 6. Rivers and main drains of Jakarta (source: NEDECO 2002)

The current situation

But, in addition to these descriptions of the historical development of the flooding problem and the 2002 flood's impact on the daily life of common people in the city, how do experts perceive the present flooding situation in Jakarta? What water management projects are under construction and what improvement measures are being taken by the government? Do experts have any comprehensive insights beyond those reported officially to explain why the flooding problems of Jakarta have not been solved in the past? Do they think that these problems can be solved, and if so, how?

Responsibility for the functioning of the drainage system lies in the hands of both central and local government. The central government (through the Ministry of KIMPRASWIL) is responsible for the river systems, whereas the local governments of DKI Jakarta and Banten and West Java Provinces are responsible for the main system of drains and the majority of local drainage works.

The drainage system of Jakarta consists of a comprehensive network of rivers, main drains, local drains, canals, retention basins, polders, flood gates, regulating gates, siphons, and pumping stations, which together regulate the discharge and retention of rainwater in relation to the Java Sea. An overview of the rivers and main drains is given in Figure 6, depicting the total water-catchment area of Jakarta.

In 1997, under the title *Study on comprehensive river water management plan Jabotabek*, at the request of the Japanese International Cooperation Agency (JICA), an extensive investigation of drainage and flood control was completed by NIKKEN Consultants and NIPPON KOEI & Co. for the government of Indonesia. In 2002 the latest study on this topic was carried out by NEDECO (2002) under the title *Final report quick reconnaissance study flood JABODETABEK 2002*. In these studies the drainage system of Jabotabek is divided into river basins. Because of their significance for the present-day comprehensive Jakarta planning approach, the most relevant of these river basins will be outlined briefly below, namely the Cisadane River Basin, Cengkareng Floodway System Basin, West Banjir Canal System Basin, the proposed East Banjir Canal System Basin, Cikarang-Bekasi-Laut (CBL) Floodway System Basin, and the Residual Basins and Urban Drainage Area in DKI Jakarta.

The Cisadane River Basin (1,411 km^2) is dominated by the Cisadane, the largest river, which has its source on the slopes of Mount Kendeng, Mount Perbakti and Mount Salak. The middle and lower reaches of the river flow through the fast-growing urban area of the city of Tangerang before discharging into the Java Sea. Water levels in the river in the city of Tangerang can reach dangerous heights. As observed in 1997, the water is sometimes dammed up by the downstream Pasar Baru Barrage built in 1937. In 1997 it

Figure 7. Photograph just upstream of Sunda Kelapa Harbour

Figure 8. Photograph just upstream of the river outlet of the Sunter

was observed that, because of a lack of proper maintenance, only five out of ten gates were operating properly. Furthermore, the backwater effects of the Cisadane River on the Sabi River are put forward as one reason for flooding in Tangerang. The Tangerang area has no urban drainage system, only some isolated channel networks.

The Cengkareng Floodway System Basin (459 km^2) is dominated by the Angke River, which has confluences with the Mookervaart Canal (connecting the Cisadane with the Angke), and the Pesanggrahan and Grogol Rivers. The system discharges into the Java Sea through the Cengkareng Drain, which was constructed in 1983 as an alternative to the extension of the West Banjir Canal proposed in the NEDECO Master Plan of 1973. The downstream reaches of these rivers flow through the densely populated area of West Jakarta. The Cengkareng Drain itself shows signs of becoming heavily silted up. As the result of land subsidence, several bridges crossing the Cengkareng Drain have very low clearances between the water level and the bridge decks.

The West Banjir Canal System Basin (421 km^2) is dominated by the 17-km-long West Banjir Canal, which was built in 1918 and collects floodwater from the Ciliwung and Krukut Rivers. It also receives water from the Cideng and Angke Drains. The Ciliwung rises on the slopes of Mount Pangrango, cutting through Bogor and the heart of Jakarta, and is a large river in terms of the size of its catchment area. In Central Jakarta, at the Manggarai Flood Gate, the Ciliwung splits, partly discharging into the West Banjir Canal and partly continuing in multiple smaller drains that discharge into the Java Sea. The Krukut joins the West Banjir Canal at the Karet Weir. In 2002 large improvement works were underway for the West Banjir Canal, such as the strengthening of embankments, dredging works, and the reconstruction of bridges (personal observation).

The East Banjir Canal System Basin (207 km^2) will be based on the East Banjir Canal that was first proposed by NEDECO in 1973 and is meant to divert run-off from the Cipinang, Sunter, Jatikramat, Buaran and Cakung Rivers along the eastern border of DKI Jakarta. At present, work on the East Banjir Canal has not yet been finished because of the high cost of land acquisition and housing compensation. Until its completion, these rivers will discharge into the Java Sea at separate locations. Under the existing situation the Cipinang joins the Sunter at Kelurahan Cipinang and continues as the Sunter, which discharges into the Java Sea at the port of Tanjung Priok. Near the eastern DKI Jakarta border, the Buaran joins the Jatikramat and continues as the Buaran. Farther downstream, the Buaran joins the Cakung and continues as the Cakung Drain. Before discharging into the Java Sea, the Cakung Drain splits into the Cakung Drain and the old Cakung River course.

The Cikarang-Bekasi-Laut (CBL) Floodway System Basin (915 km^2) is based on the 29-km-long CBL Floodway that was constructed in 1985 to divert

Figure 9. Photograph just upstream of the outlet of the Sunter River

Figure 10. Photograph just upstream of the outlet of the Cakung Drain

the run-off from the Bekasi, Cisadang and Cikarang Rivers. The main river running straight to the centre of Bekasi is the Kali Bekasi, with a catchment area of 403 km^2. The fast-growing city of Bekasi is built on the fluvial deposits of this river. Apart from some isolated channels, the urban area of Bekasi has no urban drainage system. One of the obstacles in the Bekasi River is the Bekasi Barrage, constructed in 1958, which is located in the centre of town. The situation is quite similar to the situation in Tangerang, since the barrage dams up the water during high discharges, which may lead to flooding.

The residual basins and urban drainage area of DKI Jakarta can also be seen as a catchment basin. Within the urban area, rainwater is collected by a local system of smaller drains and is discharged either directly or through pumping stations into rivers or main drains. DKI Jakarta has eighteen pumping stations for urban drainage purposes. Existing reservoirs cover approximately 145 ha and are used as retention and retarding basins or regulating ponds for pumping stations. The Pluit Retention Basin (80 ha) is the largest. Close to Halim Airport, at Sunter, a large 110-ha retention basin is planned, of which by 2002 only a part had been implemented. More and larger ponds and reservoirs are present outside DKI Jakarta in Kabupaten Bogor, Kabupaten Tangerang and Kabupaten Bekasi. Furthermore, a large number of other regulating or connecting structures, such as weirs, culverts, sluices and siphons, are part of the drainage system of DKI Jakarta.

The main point stressed by experts is that this whole system of water-catchment areas relevant to JABOTABEK has to be taken into account in order to find an answer to the Jakarta flooding problem. Until recently, in practice only limited parts of the area were taken into consideration in the execution of drainage plans. But the region has been urbanized so rapidly that no solution can be reached without taking the whole area into consideration. In the past, projects often lagged behind reality; the city may even have doubled in size between proposal and implementation of a project.

Solutions

Proposed flood solutions range from all sorts of short-term measures such as cleaning, better solid-waste management, removal of obstacles, dredging, enhancement of operation, and maintenance and improvement of infrastructure, to institutional strengthening through technical training, awareness programmes, law enforcement, and early warning and emergency assistance systems, as well as long-term improvements by means of upper watershed planning and management, and improvement of discharge and retention capacity (NEDECO 2002). The latest developments focus on extensive dredging works, eviction of illegal settlers from riverbanks, and an

Figure 11. Photograph just upstream of the outlet of the Cakung Drain

early warning system. Major plans are the massive land reclamation project called PANTURA, the East Banjir Canal currently under construction, the Ciliwung-Cisadane Tunnel, and the Cisadane-Angke Floodway.

The construction of the East Banjir Canal (EBC) was first recommended in the 1972 master plan submitted by NEDECO and elaborated by Japanese consultants in 1990. The proposed floodway has a length of 23 km and is designed to protect a large area east of the Ciliwung. EBC is planned to divert run-off from the Cipinang, Sunter, Jatikramat, Buaran and Cakung Rivers along the eastern border of DKI Jakarta. The planned floodway starts at Cipinang just northeast of Cawang and runs in an easterly direction towards Cakung, continuing north along the DKI Jakarta border, discharging into the Java Sea at Marunda. Besides the construction of the floodway, the project also includes rehabilitation works on the Sunter, Cipinang and Buaran Rivers as well as the construction of sills, river outlets, weirs, drainage outlets, roads and bridges. The total cost is estimated at Rp 5 trillion, with almost half of this budget going for land acquisition and housing compensation. At present its realization is delayed by the time required for land acquisition. The most recent planning indicates that the project can be realized by 2010. Public

opinion often considers the East Banjir Canal to be the solution to Jakarta flooding. The completion of this expensive flood control project will indeed relieve the burden caused by the rivers in East Jakarta. Caution is advised, however, in seeing this plan as a complete solution of the flooding problem. It would be more realistic to consider it simply as the completion of a thirty-year-old master plan.

Two project proposals, both still at an early stage, that have been formulated by Japanese consultants (NIKKEN Consultants and NIPPON KOEI & Co. 1997) are the Ciliwung-Cisadane Floodway and the Cisadane-Angke Floodway. The Ciliwung-Cisadane Floodway upstream from Bogor between the Cisadane and the Ciliwung is about 900 metres long and is intended to divert Ciliwung flows towards the Cisadane. The plan also includes the construction of two sediment and retention reservoirs. It aroused strong opposition from Tangerang Regency and Banten Province because of the expected environmental impact. The morphological consequences and expected discharges need to be studied in more detail. The Cisadane-Angke Floodway near the border of Tangerang and Serpong is about 4 km long and is meant to divert the water of the Angke River into the Cisadane.

The important infrastructure projects mentioned here will, if realized, result in a certain relief for Jakarta residents; however, no full guarantees can be given because urbanization of the region will undoubtedly continue to proceed rapidly, often in unpredictable or undesirable ways.

In expert reports, the burden of the flooding is usually laid on the shoulders of urban and other officials. These reports often claim that the authorities are not capable of dealing satisfactorily with all sorts of urban development issues, such as spatial planning, upper-watershed management, urban infrastructure, solid-waste management, housing and settlements, and water resources (NEDECO 2002). All sorts of financial, institutional, managerial, cooperative, participative and human-resource problems of the governmental system are considered to lie at the heart of the total inability to solve Jakarta's flooding problem. However, without denying governmental shortcomings, one can raise the question whether it is at all feasible to master all the flooding problems including urbanization with acceptable levels of investment. The answer is no. And that is why nowadays water specialists increasingly tend to propose a risk approach for dealing with flooding. Such an approach takes into consideration both the probability and the consequences of a flood event in a certain area. This means that an unpopulated area with the same prospect of flooding as a densely populated area is characterized by a much lower risk level. A plain probability approach only takes into account the prospect of flooding without reference to the material and immaterial consequences for the population involved (loss of property, human casualties, and so on).

Conclusion

From its beginnings Jakarta has been prone to flooding because of its unfavourably low location on the coast of the Java Sea. It is situated in the river basin of several rivers that transport large amounts of water and sediment during the rainy season. Although technical solutions to the flooding problem may be proposed, these require unrealistically enormous amounts of funding, while technical specialists are increasingly inclined to think in terms of risks of flooding rather than a complete solution. Engineers agree that the key to flood prevention lies in a risk-management approach, taking into account both probabilities and consequences, and that the focus in Jakarta should initially be on the rehabilitation of existing infrastructure. The problem is aggravated by rapid urbanization, inexorably demanding higher levels of water extraction leading to a steady sinking of the ground level. To raise the ground level by means of the addition of sand would lead to further sinking. That is why measures regulating spatial planning and the directions urbanization should take, should be tied in with the proposed rehabilitation. Future policies to reduce the risk of flooding in Jakarta should be based on a comprehensive water-catchment policy and should include urbanization planning. As the colonial and post-colonial anti-flooding measures of the past often strongly lagged behind the rapid growth of the city – measures often being implemented after the city population and built-up area had already expanded two or three times – the new goal should be comprehensive water and urbanization planning. Perhaps a good solution would be an elaborate system of polders. Such polders already exist in Jakarta, and during the colonial period additional proposals for polder extension were launched. A sinking city should probably be turned into a polder city. It is expected that Jakarta will have over twenty million people by 2015. Without doubt this future scenario will require large investments in flood control and flood risk management. As history shows that government agencies are often too late in implementing flood control measures, it is clear that cultural considerations and the attitude of acceptance, habituation, and disregard will not provide a strong incentive for improvement. Over the centuries Jakarta's inhabitants and government officials have become familiar with the inconveniences of floods, which are likely to be forgotten all too soon with the onset of the dry season. Moreover, flooding does not affect all parts of the city to the same extent, although in the more severe cases of flooding, even the wealthier areas fail to escape, particularly when the city grinds to a halt as we saw in 2002.

Notwithstanding the fact that at this very moment many small and large improvements in water infrastructure are in the course of being implemented in Jakarta, the overall aim should be to tackle the problem on the basis of one

comprehensive cultural, water catchment, and urbanization plan. But even then, specialists will point out that the goal to be aimed for can be no higher than the reduction of risks. Perhaps from this perspective, a completely different view of the city and its problems is required, one aimed at furnishing ample room for water. Instead of a grey or merely a green city, Jakarta should also aspire to become a blue city.

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